
Part VII

Rehabilitation and Prevention

Climbing Injury Rehabilitation

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Drs. Chris Lutter and Volker Schöffl practicing rehabilitation at "Kesslerloch", Swiss, Photo Isabelle Schöffl

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20.1 Introduction

Sports climbing imposes high load demands on the musculoskeletal system, and periodically climbers, like all sports participants, will sustain injury. The evidence base in regard of the aetiology and pathology of climbing-related injuries is increasing, but currently there is a paucity of published research on rehabilitation [1]. In recent years, the rise in the popularity of sport climbing means it is increasingly likely that healthcare professionals will be required to diagnose and treat climbing-related injuries. It is therefore vital that those involved in the care of climbers have an understanding of the principles of rehabilitation [2] and an explicit knowledge of climbing-related injuries and their management [3].

20.2 Rehabilitation and Physical Preparation

Rehabilitation is the enhancement of tissue, functional, emotional and psychological function after injury to structure and/or impairment in physiology [4]. It is a planned and systematic approach to the judicious application of physical load for the restoration of function [5]. However, this definition may be too restrictive for the athlete as it does not fully address the high-performance outcomes sought from a rehabilitation programme. The philosophy must include both a rehabilitation and a physical preparation mindset in which the performance outcome is the key driver for all decision-making [6].

Rehabilitation for use in the general population is focused on the restoration of basic function such as activities of daily living. This does not adequately address the high functioning athlete and successful return to sport [7]. Therefore, levels of physical restoration must go beyond traditional rehabilitation paradigms if the athlete is to return to climbing in optimal condition [8]. The principles of rehabilitation and physical preparation are influenced by tissue healing time frames. The early-to-middle stages of recovery are often characterised by low tissue loading,

restoration of optimal joint range of motion, soft tissue mobility and the re-establishment of motor control and movement patterns [9]. In contrast, the late phases and return to climbing are the physical preparation stages. Its focus is on the development of high load/force interventions to improve physical qualities, tissue capacity, technical skill and tactical abilities necessary for optimal climbing performance [10].

20.3 Principles of Management

Rehabilitation for the climber regardless of the actual injury should be based upon evidence-informed principles. At the core of any management, intervention is the judicious application of appropriate research, the clinician's experience and shared decision-making with the patient [11]. This awareness of the need for a balanced appraisal of practice evidence is critical for situations where empirical research is limited which is the current situation in climbing-specific rehabilitation [3].

20.3.1 Early Stage

The principles of management can be characterised into anatomically specific interventions and regional and global interventions [12]. Anatomically specific interventions involve management of local impairments and pathology specific to the injury. This is often associated with the early to middle stages of the climber's rehabilitation. For example, interventions that improve the material and morphological properties of isolated muscular, tendon and connective tissues occur at this stage [13].

Injury is often associated with a period of deconditioning and disuse because of pain and immobilisation [7, 14, 15]. It is well documented that lengthy periods of immobilisation can lead to reduced muscle strength and impaired load tolerance of connective and tendon tissues [16, 17]. The restoration of optimal joint and soft tissue mobility is important to establish early during the rehabilitation of the climbing athlete. Appropriate

joint and soft tissues mobility is important from a biomechanical perspective. Optimal joint mobility provides a basis for force generation, force transfer and force absorption [18, 19]. Suboptimal mobility prevents the climber from obtaining the most efficient body shape to facilitate optimal biomechanics. The integration of kinetic chain training early in the climber's rehabilitation is of critical importance [20]. Climbing is a whole-body strength and power sport reliant on upper and lower body interactions for optimal performance [21, 22].

The trunk musculature [23, 24] and lower body [18] has been shown to provide a strong base for force production and kinetic energy transfer from the ground to the upper extremity in non-climbing studies. Anecdotally, many climbing coaches will attest to the importance of trunk muscle force production for optimal performance during climbing (see Fig. 20.1). The case for training the hip musculature and lower limb strength is more controversial as this is not as well recognised by climbing coaches for influencing climbing performance. Research from other upper limb-based sports increasingly recognises the importance of developing appropriate strength and power development in the lower limb as a basis for athletic performance [15, 20, 25].

20.3.2 Middle-Late Stage

Regional- and global-specific interventions which increase kinetic chain integration are associated with the middle to latter stages of rehabilitation and physical preparation [26]. At this stage, the management aims are twofold:

1. To increase sport relevant physical capacities
2. To improve and/or maintain climbing skill acquisition and tactical awareness

The improvement in sport relevant physical capacities such as maximal muscle strength, explosive strength (rate of force development) are not sports specific but are sport relevant to climbing [27, 28]. This type of training is best achieved with mainly non-climbing-based activities that



Fig. 20.1 Retraining trunk, shoulder and lower limb integration



Fig. 20.2 Skill-related training at the physical preparation stage

can quantifiably overload specific muscle groups without the complication of the climbing wall. It involves the development of generic physical qualities using traditional resistance equipment that form the foundation for sports-specific climbing, skill acquisition and tactical development. It is the expression of these physical qualities in a sports-specific situation which allows for optimal biomechanical movement patterns and climbing performance (see Fig. 20.2). These generic capacities form the basis of a physical preparation programme for full return to unre-

stricted climbing at a high level of performance [29, 30].

20.4 Other Professionals Allied to Performance

Optimal management of the climbing athlete requires a multi-disciplinary team approach [31]. The rehabilitation professional must understand their scope of practice in terms of knowledge, expertise and their limitations. This may often mean collaborative working, and communication with other professionals allied to the climber as required to the climber is required. The climber must always be at the centre of the decision-making process may be coordinated by a designated performance manager. Therefore, an understanding of when to initiate onward referrals, for example, to coaching staff, dietitians, surgeons, strength and conditioners and podiatry is important for optimal performance management [32].

20.5 Exercise and Mechanical Loading

The management of mechanical load on the human musculoskeletal system is one the fundamental tenants of effective rehabilitation in athletic populations [32, 33]. Mechanical forces applied to biological tissues has been termed mechanotransduction [34]. The application of mechanical load on biological tissues has been shown to facilitate adaptations in the morphological, material and neurological properties of tissues [35]. In general, many of the physical therapies including soft tissue massage, electrotherapeutics, exercise therapy and strength and conditioning modalities share commonalities with the principle of mechanotransduction [36]. The commonality shared by these approaches is the application of load to musculoskeletal tissues facilitates changes at cellular, organ and system level [37]. For example, the use of electrotherapies such as shockwave therapy or pulsed ultrasound in the early phases of rehabilitation involve the application of mechanical load to tissues at

the cellular level. There is preliminary evidence to suggest adaptation in tissue morphology at cellular level with various electrotherapies. Connective tissues have been shown to sense mechanical stimuli via signalling molecules in the extracellular matrix (ECM) and cytoskeletal structures of cells [38].

At the cellular level, mechanical forces tension the adherent ECM which then signal mechanical receptors via the integrin. It has been shown that stimulation of the integrin receptors in cultured cells respond to mechanical load by increasing focal adhesions [39]. This may be an active adaptation to mechanical load promoting cellular growth, survival and morphological changes. For example, when the integrin is stimulated experimentally with mechanical forces, the cellular mechanisms can be seen to realign along the lines of stress or applied tensional loads [40]. While this theoretical perspective may appear far removed from the climbing athlete in clinical practice, it is noteworthy that this underlies and supports all the higher-level adaptations at tissue, organ and system levels and hence their performance outcome. At the tissue and organ levels, changes in morphology, physiology and material properties of muscular and connective tissues can be achieved by long-term programming of the climbing athlete based upon mechanical loading [41–43]. For example, the cornerstone of physical rehabilitation and physical preparation programmes is the improvement of bio-motor capacities such as strength, endurance, power and rate of force development in muscular and connective tissues [44, 45].

20.6 Blood Flow Restriction

In recent years, adjuncts such as blood flow restriction (BFR) have shown promising results in clinical populations with high-quality studies [40, 46].

Blood flow restriction training has been shown to produce significantly improved gains in muscle strength and hypertrophy in injured load-compromised populations [47]. The underlying mechanism of action is not known for sure,

but the occlusion of blood proximal to the muscle belly is thought to increase fibre type recruitment, metabolic accumulation, cellular swelling and stimulation of protein synthesis [48]. This modality shows promising results in that high intensity muscular adaptations can be achieved with very modest training loads [49]. For example, the American college of sports medicine (ACSM) recommend load intensities of 70–80% of the one repetition maximum (1RM) to achieve adaptations in muscular strength, power and hypertrophy [27]. However, this type of load is often contraindicated in an injured population because of pain and soft tissue insufficiency [29].

BFR has been consistently shown to elicit hypertrophic adaptations in muscular tissues with training intensities as low as 20% of 1RM [50]. This has several advantages for the climbing athlete during rehabilitation. Firstly, it allows the early therapeutic application of load in contractile tissues that either cross or are closely associated with load-compromised structures such as the radioulnar joint or annular pulley ligaments [51]. For example, after sustaining an annular pulley injury, a climber could potentially begin resisted finger exercises earlier in their rehabilitation. This would minimise the risk of applying excessive mechanical loads through the healing tissues while still maintaining muscle strength in the forearm flexors. Secondly, it theoretically allows the training of non-injured regions to a high level without compromising injured areas [52]. For example, hang board training or other specific training modalities could be utilised to overload the finger and forearm muscles without compromising a load-impaired shoulder joint.

20.7 Electrophysical Therapy and Other Therapeutic Adjuncts

20.7.1 Electrophysical Therapy

Electrophysical therapies have been long established in clinical practice as adjuncts or standalone interventions for the management and/or rehabilitation of injuries and disease.

Electrophysical therapies are used for symptomatic relief of pain and discomfort, to reduce muscle spasm, to resolve inflammation and to facilitate tissue healing.

During electrophysical therapy, apparatus (devices and accessories) is used to deliver energy (stimuli) into the body to provoke physiological responses at cellular, tissue, organic and whole-body levels to maintain and optimise health [International Society for Electrophysical Agents in Physical Therapy (ISEAPT) <https://www.wcpt.org/iseapt>]. Energy is delivered across the intact surface of the skin via conduction, convection, conversion, radiation or evaporation to alter physiological processes in ‘target tissue’ for therapeutic benefit. Physiological actions include alterations in the structure and/or activity of substrates involved in metabolism and healing (enzymes and proteins) or the modulation of neural activity. Despite much research, there is insufficient robust clinical research evidence from randomised controlled clinical trials to make confident judgements about the effectiveness of most electrophysical therapies.

There are many types of therapeutic device available to practitioners and most have pre-programmed menu-driven settings that manufacturers claim to be beneficial for a bewildering array of medical conditions, often leaving practitioners confused. The principles of different types of electrophysical techniques have largely remained unchanged over time. Different types of electrophysical therapies can be categorised as follows:

Thermal and microthermal

- Superficial heating using contact or radiant techniques
- Pulsed and continuous shortwave and radio-frequency therapies
- Cooling (cryotherapy) using contact and radiant heat

Nonthermal, microthermal and light energies

- Therapeutic ultrasound (US)
- Low level laser therapy (LLLT)
- Ultraviolet therapy



Fig. 20.3 A therapist using extracorporeal shockwave therapy



Fig. 20.4 Electrical stimulation used to superimpose muscle contractions on normal movement patterns for motor relearning

- Magnetic and pulsed magnetic (PEMF) therapies
- Extracorporeal shockwave therapy (ESWT) (Fig. 20.3)
- Vibration

Electrical stimulation

- Transcutaneous electrical nerve stimulation (TENS)
- Neuromuscular electrical stimulation (Fig. 20.4)
- Functional electrical stimulation
- Alternating currents including interferential therapy (IFT) and Russian stimulation
- Electrical stimulation for wounds

Considerations for practitioners include selection of electrophysical modality and device, target tissue, energy parameters (device settings), dose and regimen. Failure to dose within the therapeutic window may result in harmful (toxic) or subtherapeutic effects. An electrophysical therapy may produce different physiological actions by altering the parameters and dose of the energy. If used appropriately, electrophysical therapies are considered less harmful than medical interventions such as

drugs, especially to provide symptomatic relief. Watson and Nussbaum [53] and the International Society for Electrophysical Agents in Physical Therapy [54] provides comprehensive evidence-based coverage of the main electrophysical therapies used in clinical practice.

Professional bodies recommend only a minority of electrophysical therapies as adjuncts to core treatment for specific medical conditions (e.g. TENS for osteoarthritis and rheumatoid arthritis). Hypothetically, PSWT, US, LLLT and electrical currents may increase the rate of division of malignant tissue and for other rapidly dividing tissue (embryo/foetus), so they are not applied over areas of malignancy in treatable cancer. Generally, a history of malignancy is not a contraindication. Pregnancy is not a contraindication provided that energy is not applied close to embryonic/foetal tissue. Nevertheless, if used appropriately, electrophysical therapies are considered less harmful than medical interventions such as drugs, especially to provide symptomatic relief. A comprehensive resource of contraindications and precautions have been reported previously [55].

The principles of treatment are to assess contraindications and appropriateness of treatment

modality for the patient and to seek informed consent. Further considerations include:

- **Technique:** The sequence of actions necessary to administer the treatment
- **Dose:** Quantity of treatment to be administered at one time
- **Dosage:** Size (amount/concentration), frequency and number of doses of the treatment (per treatment session)
- **Regimen:** A program of treatment sessions, e.g. the course of treatment, how many sessions and how often

20.7.2 Extracorporeal Shockwave Therapy

Extracorporeal shockwave therapy (ESWT) is a therapy that has recently gained popularity in tendon-related disorders in clinical practice and within the literature.

ESWT is an electrotherapeutic modality that uses acoustic wave formations to deposit high energy into musculoskeletal tissues (see Fig. 20.5) [56]. It is a form of mechanotransduction which aids the regeneration and reparative process of bones, tendons and connective tissues [57]. ESWT is an increasingly popular therapeutic modality in the management of tendinopathy and connective tissues disorders [58]. From a physiological stand point, the acoustic waves utilise both positive and negative phases of the shockwave. The positive phase creates direct mechanical forces, and the negative phase generates high speed cavitation's [41]. The proposed mechanisms of action include the promotion of neovascularisation at the tendon bone interface, the stimulation of tenocytes, increased leukocyte infiltration and the proliferation of growth factor and protein synthesis [56]. This may stimulate collagen synthesis and tissue remodelling [57].

Electrophysical therapies may be beneficial when used as an adjunct to core treatment or within a package of care. For example, electrical stimulation therapies (see Fig. 20.4) can be

used to prevent muscle atrophy after injury [42] or to superimpose muscle contractions on normal movement patterns for motor relearning [37]. These modalities likely yield the most benefit during the early stages of rehabilitation when musculoskeletal tissues may be load compromised due to pain and/or effusion and/or immobilisation [38, 39]. However, as the athlete progresses through the stages of rehabilitation and physical preparation, these modalities become less important, and exercise-based loading regimes take on greater significance.

Glossary of terms: electrical qualities

Current	Flow of positive electric charge. The strength of current flow in any medium is related to voltage differences in that medium, as well as the electrical properties of the medium, and is measured in amperes
Voltage	Measure of the difference in electric potential between two points in space, a material, or an electric circuit, expressed in volts
Amp	Unit of electric current. One ampere corresponds to a certain number of electrons passing a fixed point each second
Resistance	Opposition offered by a body to the passage through it of a steady electric current
Impedance	Measure of the total opposition to current flow in an alternating current circuit, made up of two components, ohmic resistance and reactance
Reactance	Opposition to the flow of alternating current caused by the inductance and capacitance in a circuit rather than by resistance
Inductance	Property in an electrical circuit where a change in the electric current through that circuit induces an electromotive force (EMF) that opposes the change in current
Capacitance	Measure of the ability of a configuration of materials to store electric charge. In a capacitor, capacitance depends on the size of the plates, the type of insulator and the amount of space between the plates
Wavelength	Distance between one peak or crest of a wave of light, heat or other energy and the next corresponding peak or crest

Frequency	Number of crests of a wave that move past a given point in a given unit of time. The most common unit of frequency is the hertz (Hz), corresponding to one crest per second. The frequency of a wave can be calculated by dividing the speed of the wave by the wavelength. Thus, in the electromagnetic spectrum, the wavelengths decrease as the frequencies increase and vice versa. More simply the number of occurrences of an event per unit time
Intensity	Magnitude, as of energy or a force per unit of area, volume, time
Pulse frequency	Pulses per second
Pulse duration/width	Length of time for each pulse
Pulse shape (waveform)	Refers to the build-up and reduction in intensity during a single pulse

20.7.3 Manual Therapies

Manual therapies are approaches and techniques that involve the skilled use of the hands to rub, knead, mobilise and manipulate musculoskeletal and neural tissue in and around joints [33]. Manual therapies are primarily used to treat conditions affecting musculoskeletal structures. They do this by increasing tissue extensibility and decreasing muscle tone and tension which improves range of motion at joints and movements that are restrictive. Manual therapies are also used to reduce swelling, inflammation and pain [59]. There are basic physiological principles underpinning manual therapy and diversity in philosophies, concepts, approaches and techniques according to specialism.

There is a paucity of robust research evidence to judge whether there are differences in benefit and harm between manual therapy specialisms or between different approaches and techniques within specialisms. Consequently, practitioners often base clinical decisions on their previous experience, resulting in inconsistencies in approaches and techniques selected for patients with similar clinical presentations.

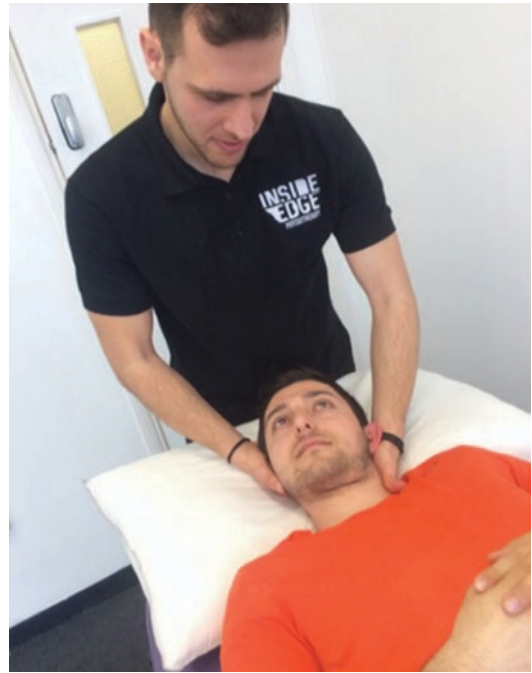


Fig. 20.5 A therapist using spinal manipulation as a treatment modality

20.7.4 Joint Mobilisation and Manipulation

Joint mobilisation and manipulation are manual therapy techniques used in isolation or in combination to restore normal function at spinal and peripheral joints.

Joint mobilisations are active and/or passive physiological movements and/or low velocity passive accessory movements of varying amplitude within or at the end of range of motion and are used primarily to restore movement by inducing sympathetic nervous system excitation [60]. Joint manipulations (see Fig. 20.5) are high velocity, low amplitude, thrust techniques, within or at the end of range of motion and are used primarily to restore movement by inducing cavitation and the separation of articular surfaces at the joint [33].

Treatment effects are broad, may be immediate and/or long-term and include reduction in pain, lengthening of periarticular tissue, increase range of motion and reduced joint stiffness.

Contraindications and precautions should be checked prior to treatment and include fractures, dislocations, cauda equina, rheumatological conditions, osteoporosis and vertebral artery insufficiency (cervical mobilisations). Joint mobilisation is considered a safer adjunct to core treatment. Potential risks are generally considered to be low and include increased pain, proximal and/or distal to the treatment site [61]. A variety of mechanisms of action have been offered to explain the therapeutic outcomes for joint mobilisations. In 2020, a systematic review and meta-analysis of 17 studies provided evidence that joint mobilisation elicited sympathetic-excitatory effects [62], such effects may reduce pain and stiffness.

20.7.5 Evidence for Clinical Efficacy

There is a paucity of robust clinical evidence to establish optimal technique for joint mobilisations and manipulations. Thus, in clinical practice, techniques vary considerably according to the background and previous clinical experience of the practitioner and poses challenges when deciding appropriate techniques to use in randomised controlled clinical trials, and when synthesising findings from previous research. In 2013, a systematic review of 22 studies found moderate evidence to support the use of joint mobilisations for lateral epicondylalgia and insufficient evidence to support use in conditions of the wrist and hand [63]. In 2019, a meta-analysis of seven studies provided strong evidence that joint mobilisations improve pain and/or functional grip in people with lateral epicondyle tendinopathy [64].

20.7.6 Summary

Joint mobilisations and manipulations are often used in conjunction with other treatments; it is therefore difficult to ascertain the effect of mobilisations per se. Nevertheless, evidence supports consideration of their use as an adjunct therapy to reduce pain and increase range of motion in musculoskeletal conditions.

20.8 Specific Conditions

20.8.1 Lower Limb

20.8.1.1 Acute Hamstring Injury

This is often associated with a heel hook mechanism of injury and is characterised by an over-strain of the contractile fibres of the hamstring muscle group [65]. This can include the bicep femoris, semimembranosus and semi tendinosis muscles [66]. The mechanism of injury is often an acute presentation with the climber complaining of a sudden onset of pain and weakness during the heel hook technique [67]. This hamstring injury is unique in that the mechanism of injury is a concentric muscle action [68] which contrasts the eccentric mechanism of injury often seen in high speed sprinting and team sports [69].

20.8.1.2 Initial Management

Initial management involves the standard protection, rest, ice, compression and elevation protocol (PRICE) (see Table 20.1) [70]. The injured muscle group can be taped or splinted into a shorten position to avoid excessive tensional forces during the first 1–5 days dependent upon the grade of injury [71]. The injury severity is generally graded from grade 1–3. Grade 1 is associated with minimal

Table 20.1 PRICE protocol [70]

Protection	Protection of the injured area from further insult avoids healing delay
Rest	Relative rest of the injured area is encouraged to allow healing. This includes avoidance of activities that are stressful to the injured area. Complete rest is often to be avoided as controlled pain-free mobilization of injured tissues can aid recovery
Ice	Cooling reduces local tissue metabolism and limits the extent of injury. This may reduce secondary cell necrosis. Cooling of tissues may provide pain modulation via the active pain gate mechanism
Compression	Compression of the injured area helps to reduce swelling and oedema
Elevation	Elevation of the injured area reduces blood flow to the injured area and reduces swelling

disruption of the contractile fibres and connective tissues, grade 2 is associated with more significant disruption of the contractile and connective tissues, and grade 3 is associated with catastrophic failure of the tissues leading to complete rupture [14].

20.8.1.3 Definitive Management

The cornerstone of optimal management for an acute hamstring injury is a progressive tissue strengthening programme [68, 72]. The programme should aim to address the force-generating capacity of the muscle with a bias towards concentric muscle actions. In the initial phase of management, strength training of the injured tissues can begin with isometric exercises but should rapidly progress to concentric exercise as pain, irritability and tissue healing allow [69]. In the early stages, when tissues are load compromised, adjuncts such as blood flow restriction can be used to facilitate high intensity muscular adaptations with modest training loads [49]. In the latter stages of the rehabilitation programme, maximal strength in both isometric and concentric muscle actions take priority (Fig. 20.6). The programming of high velocity strength training will be specific to the performance requirements of the climber. For example, there may be a greater performance requirement for high velocity strength training for a boulder and speed climber than a lead climber [70, 71]. At the return to climbing stage, the athlete should have no pain or hesitancy during the heel hook manoeuvre in various climbing-specific positions. Additionally, there should be no more than a 5% deficit or less in concentric muscle strength between the affected and contralateral side [29, 72].



Fig. 20.6 Single leg hamstring muscle strengthening with core muscle integration

20.8.2 Ankle Sprains

Injury to the ligaments of the ankle and foot is a common cause of chronic disability and recurrent instability in athletes [73]. Sprains of the lateral ankle ligamentous complex involving the anterior talofibular ligament, +/- the lateral and posterior calcaneofibular ligaments are often injured when the ankle is inverted and planter flexed at the time of injury [74].

This injury often involves a high-energy mechanism of injury in all but a few cases. The mechanism of injury is normally associated with a fall from height when bouldering [75] or an uncontrolled fall during sport or traditional climbing [76]. The injury mechanics classically involve inversion for the midtarsal and subtalar joints and planter flexion of the talocrural joint.

The ankle complex is unique in its anatomical configuration in that the shape of its bones are irregular conforming to very precise anatomical relationships. This provides a friction-free environment for weight bearing locomotion [77]. The ligamentous structures act as somatosensory force transducers to control precise joint motion. Injury to the ligaments affects the mechanical properties of the collagenous tissues and also the mechanoreceptors responsible for conveying information, influencing joint position sense and kinaesthesia to the central nervous system [78]. Poor initial management and rehabilitation of ankle sprains can lead to pain, disability and functional instability [79]. Chronic functional ankle instability is associated with lower satisfaction, lower rates of return to sports participation and increased referral rates for surgical opinion [80]. Therefore, we believe that appropriate initial management of ankle ligamentous injuries and subsequent rehabilitation is critical for the functional restoration of the climbing athlete.

20.8.2.1 Initial Management

The initial management of an ankle sprain should follow the principles of PRICE to prevent further insult to the injured tissues and to create optimal conditions for the restoration of tissue homeostasis. Appropriate ankle bracing or taping to mini-

mise mechanical directional stress on the injured ligamentous structures has been shown to promote early mobilisation [81].

20.8.2.2 Definitive Management

The management approach should include a comprehensive rehabilitation programme which addresses all deficits in range of motion, muscle strength and proprioceptive control [82]. Appropriate range of motion into both dorsal flexion and planter flexion is important for shock absorption during landings and for propulsion during climbing. Failure to restore functional dorsal flexion range of motion may inhibit the transfer of momentum [79] during activities such as rockovers and prevent optimal mechanics during activities such as Egyptians and frog positions. Ankle planter flexion is part of the coordinated system of lower quadrant triple extension for dynamic propulsive activities such as dynos [83]. Failure to restore this movement range of motion frequently leads to reduced ability to overcome the effects of gravity such as is the case when executing dynamic movements.

The restoration of balanced strength development is also important from a force absorption and force generation perspective [12]. Pain inhibition and disuse is a frequent cause of impaired muscle performance in the injured climber. Often this impairment is exacerbated by inappropriate muscle conditioning and too rapid a return to sports participation [84]. Therefore, we recommend that muscle strength of the planter flexor, dorsal flexor, inversion and eversion muscle groups be within 5% of the contralateral side before return to unrestricted climbing. Ankle sensory motor retraining is also important after injury to ligamentous structures (see Fig. 20.7). This is most important for optimal foot posture during landing to prevent reinjury [85] and for effective and precise footwork when climbing. Therefore, consideration should be given to progressive training of landing technique on stable and then unstable surfaces [80]. This progression should take account of standard rehabilitation progressions involving landing from low heights to progressively higher heights [78]. This will increase the ground reaction forces and load



Fig. 20.7 Climber participating in ankle sensory motor retraining

tolerance through the ankle complex in a progressive manner. Consideration should also be given to skill-related training to develop sensory motor control of precise footwork during simulated conditions and while climbing. All rehabilitation systems should be process driven with therapeutic exercise progressing from low load to high load, simple movements to complex movements and small ranges of motion to large ranges of motion [10].

20.8.3 Upper Limb

20.8.3.1 The Painful Wrist

The wrist is a common area of pain and dysfunction in climbing populations [76]. Wrist pain in climbers is often situated around the radiocarpal joint and typified by a chronic aetiology of impaired functional capacity during mechanical load [86]. In chronic wrist pain, it is often difficult

to identify a specific anatomical target to focus rehabilitation interventions [87]. Therefore, local management of the chronically painful wrist is unlikely to change the systemic load transferred and absorbed by the complex during climbing. It is often more pragmatic for the clinician to investigate why the wrist complex has failed to manage the applied loads imposed by climbing [88].

20.8.3.2 Definitive Management

Local management of pain-sensitive structures should be instituted to reduce the effect of pain inhibition and localised inflammation if that is present. Standard procedures utilising PRICE and nonsteroidal anti-inflammatories may have a place in the initial management of the climber with persistent wrist pain [88]. In addition, the climbers training and competition volumes should be modified to unload the pain-sensitive and load-intolerant structures [26].

20.8.3.3 Local Assessment

A local assessment of the wrist should include the following:

1. Palpation of the radioulnar joint lines and associated palmar and volar ligaments
2. Palpation of the carpal bones and associated ligaments
3. Assessment of muscle performance of the following:
 - Flexor carpi radialis
 - Flexor carpi ulnaris
 - Extensor carpi radialis longus and brevis
 - Extensor carpi ulnaris
 - Abductor pollicis longus
 - Abductor digiti minimi
4. Distal and proximal radioulnar joint accessory movements
5. Joint position sense in low and high load situations

20.8.3.4 Global Assessment

A comprehensive assessment of the climber's technique on the wall should be examined to establish its relationship to movement efficiency. This can be done by the therapist depended upon their expertise or in conjunction with a climbing

coach. The practitioner is looking for evidence of movement inefficiency that might be contributing to excessive load through the hand and wrist complex [33], refer to Chap. 21.

The neuromuscular assessment should examine the strength of the shoulder girdle in particular the rotator cuff muscles and axial scapula adductors and retractors. A general assessment of static shoulder girdle and trunk posture should be undertaken to establish muscular balance between the anterior and posterior upper body musculature [57]. We have noted an anecdotal relationship between the poor muscular development of the shoulder girdle abductors muscles when compared to the shoulder adductors and internal rotators in climbers with chronic wrist and elbow pain.

The rotational profile of the proximal and distal radioulnar joints should also be assessed for total range of motion into supination and pronation. Loss of range of motion and excessive soft tissue tightness can lead to impaired function of the wrist complex during climbing. The loss of pronation range of motion can theoretically lead to increased shear forces at the wrist complex. This is because the radiocarpal joint attempts to compensate for the reduced rotatory range of motion at the proximal and distal radio ulna joints [89]. The radiocarpal joint has two degrees of freedom: flexion and extension and ulna and radial deviation. Rotation occurs functionally at the superior and distal radioulnar joints [90]. Therefore, assessment of soft tissue compliance at bicep brachialis and supinator muscles is indicated in addition to the accessory movement of the proximal and distal radioulnar joints.

20.8.3.5 Rehabilitation

The rehabilitation programme should be prescriptive based upon the impairments found after a comprehensive evaluation of the climber's wrist, hand and upper quadrant. Rehabilitation should then aim to restore optimal function where this can be extrapolated to load intolerance of the wrist complex (Fig. 20.8). There is often an underlying cause for tissue incompetence and persistent symptoms, for example, poor joint position sense or reduced proximal radioulnar range of motion.



Fig. 20.8 The restoration of mechanical wrist function using resistive loading

Significant impairments must be addressed based upon the principles of rehabilitation and physical preparation already discussed to ensure optimal management and return to climbing.

20.8.4 Closed Traumatic Flexor Tendon Pulley Rupture

Injuries to the annular flexor pulley ligaments are one the most common injuries to afflict climbing populations [91]. This type of injury is particularly debilitating because of the significant impairment in climbing performance associated with rupture [3]. The pulley ligaments are connective tissue structures composed primarily of type 1 collagen [92]. The pulley ligament provides a critical biomechanical role ensuring optimal force transfer from the strong flexor tendons to the finger and climb-

ing interface [93]. Anatomically, the flexor pulley ligaments are extracapsular thickenings of the joint capsule and are supported by the cruciate ligaments laterally [83]. There are five annular pulleys which alongside the cruciate ligaments and fibrous capsular thickening act to constrain the movement trajectory of the long flexor tendons [94].

20.8.4.1 Initial Management

The initial management regardless of the severity of the injury will follow the standard principle of PRICE [70]. The management of pain and the prevention of further soft tissue insult is the primary objective in the first 24–48 h after the injury. The suppression of inflammation at the onset of injury is less important and only becomes a priority if it continues beyond an acceptable time frame [95]. The protection of injured structures with rest and appropriate splinting [96] or taping [97] is helpful to facilitate the approximation of the bone tendon interface [96]. This theoretically assists with scar tissue formation and helps to prevent excessive bowstringing of the tendon which may delay healing. The number of pulley ligaments injured and the degree of disability will determine whether surgical or conservative management is chosen [98].

20.8.4.2 Definitive Management

The management process for rehabilitation of the partial or ruptured pulley ligament follows the principle of progressive overload [95]. In the initial stages, mechanical load can be administered with electrotherapeutic modalities such as ultrasound and laser therapies [34]. However, the cornerstone of load management is active resistance training [16, 41, 99]. This should be applied judiciously in the early stages of tissue healing (see Table 20.2) with resistance training to isolate load to the digit of the injured pulley. Loading progressions can start with active flexion at the proximal interphalangeal joint (IPJ), middle IPJ and then distal IPJ. Resistance training bands are a pragmatic method to titrate mechanical load to the recovering pulley ligament via the injured digits. Light and then eventually moderate to heavy resistance bands can be progressively added as tissue healing progresses. In the early stages of rehabilitation climbing should not be permitted

Table 20.2 Stages of soft tissue healing (Adapted from Jimenez and Jimenez (2004) and Watson (2008) [100, 101])

Phase	Description	Time course	Interventions
Bleeding phase	This phase occurs after the initial injury and is characterised by an increase in vascularity at the site of injury	Start 0–1 h Peak re-activity 4–6 h	PRICE interventions
Inflammatory phase	The inflammatory phase is a necessary component of soft tissue healing. It is composed of both a vascular and cellular cascade which have direct effects on the tissue repair sequence	Start 0–4 h Peak re-activity 1–3 days	PRICE interventions
Proliferative phase	This phase begins with the formation of granulation tissue which eventually matures to become scar tissue. The catalyst for this phase is via cytokine mediators from the inflammatory phase	Start 24 h Peak re-activity 2–3 weeks End 4–6 months	Low to higher resistive exercise loads
Remodelling phase	The remodelling phase is characterised by the refinement of the collagen from immature to more robust tissue. The deposition of collagen fibrils with appropriate loading contributes to orientational alignment along the lines of stress	Start 24 h Peak re-activity not known End 12 months	Moderate to high resistive exercise loads

[102]. The maintenance of soft tissue mobility to promote an extensible scar and fibroblastic reorganisation along the lines of mechanical stress can be achieved with mobility interventions such as tendon gliding exercises [103].

In accordance with Wolff's law of mechanical loading and adaptation, resistance loads should be increased for continued adaptation [16]. Loading stimulates and increases the morphological and material properties of the connective tissue [16], muscle [104] and tendon [105]. Although this has not been shown directly *in vivo*, there is substantial evidence from animal studies [106], *in vivo* tendon studies [107] and cross-sectional studies [108] in competitive climbers which suggest that mechanical load induces morphological changes in connective tissues such as ligaments.

Therefore, the aim of rehabilitation and physical preparation is the restoration of tissue strength to preinjury levels. As the stage of tissue healing progresses from the proliferative to the remodelling stages, the degree of loading can increase to include training modalities which apply greater concentrated loads to the digits with fingerboards utilising holds with progressively smaller contact areas [108, 109] (Fig. 20.9). This will thereby



Fig. 20.9 Training modality with concentrated mechanical loading for the fingers

encourage further tissue adaptation. Resistance bands can eventually be replaced with traditional weighted pulley systems that stress the injured digit in insolation and collectively.

Participation in climbing-specific activities should be reserved for the intermediate stages of the climber's physical preparation programme. It is often tempting to return to climbing early with an open grip hand position because the pulley tendons (A2 and A4) are loaded to a significantly lesser degree than the full crimp position [110]. This is often the anecdotal rationale for early return to climbing using an open grip. However, this strategy is problematic.

The A2 and A4 pulleys are at most risk of injury because of their larger internal moment arms compared to the A3, A1 and A5 pulleys ligaments [111]. Therefore, although under less load during open hand grips, the A2 and A4 are still vulnerable in the early stages of rehabilitation. It is also practically difficult to quantify tendon and pulley mechanical loading during climbing because of the various configurations of hold types [105]. We therefore suggest it is safer to restrict climbing until the injured digit can be loaded with no symptomatic reaction during four finger isolated finger strengthening in a half crimp position. This reduces the risk of over training and reinjury and ensures a degree of tissue strength [88, 94]. Return to climbing too early increases the risk of overzealous risk taking which could cause accidents such as foot slips and excessive application of loads when negotiating difficult climbing problems [102].

Progressive resistance training principles should be followed [32]. We recommend isolated digit strengthening proceed climbing specific activities. The injured digit should be loaded in insolation starting with isometric muscle actions, progressing to concentric actions and then finally to eccentric actions [12]. Each muscle actions must be accompanied with symptom-free participation of the affected digit before progression to the next muscle action [10]. This rationale is borne out by the high mechanical forces attributed to eccentric loads in the contemporary literature [112, 113] which may predispose the ligamentous tissues to excessive load if used prematurely.

The resumption of climbing should begin when the affected digit is able to tolerate concentric forces within 5% of the contralateral digit. The range of motion along the course of the flexor tendons in both flexion and extension should be full range and asymptomatic. On palpation of the soft tissues overlying the pulley ligaments this region should not reproduce the patient's original symptoms. This return to sport criteria is in accordance with other published literature for other regions of the body [29, 114, 115].

The resumption of climbing is a phased approach with a progressive increase in volume load over several weeks and months depending upon the grade of injury. In the initial phases, the duration of climbing, the grade and number of hold types is limited. For example, in the first 1–2 weeks of climbing practice, the climber may only climb 2× per week restricted to only an open grip, accumulating up to a maximum of 100–150 isolated hand grips in each session. At week 5–6, the frequency of climbing may remain the same, but the climber is allowed to use half crimp or open crimp holds in addition to open grip holds accumulating between 150 and 200 isolated hand grips during a session. As the rehabilitation and physical preparation progresses, there will be a progressive increase in the volume of climbing each week, the use of full crimp holds, duration of each climbing session and inclusion of steeper over hanging terrain. The cardinal rule which must be enforced during this phase is that climbing at each stage must be symptom free.

20.9 Shoulder

20.9.1 Chronic Subacromial Pain Syndromes

The shoulder complex is often a common source of pain and dysfunction in climbers. Traditionally, pain within the shoulder complex has been managed from an anatomical perspective. This aims to establish which of many pain-sensitive structures is the source of the patient's symptoms. While this may be possible with a traumatic injury, this is often not possible in cases of per-

sistent shoulder pain [116]. It is difficult to differentiate which structures are at fault with physical examination tests because of their low diagnostic accuracy [117]. Most physical examination tests for the shoulder complex are pain provocation tests. This renders it difficult to obtain a structurally specific diagnosis [118]. Other evidence supportive of this position is diagnostic imaging studies. Research from imaging interventions such as magnetic resonance imaging [119] and ultrasound sound studies [120] often report incidental findings in asymptomatic populations.

In patients with chronic persistent shoulder pain, it is highly probable that several structures will be symptomatic acting in combination as a source because of the close anatomical relations of pain-sensitive structures such as bursa, tendons, capsule and joint articular surfaces [121]. For example, the supraspinatus tendon, bicipital tendon and hypertrophied subacromial bursa can all be symptomatic during common climbing positions which require shoulder elevation. In respect to this difficulty, treatment approaches based upon specific tissue pathology and structural diagnosis may be therefore problematic [122].

Rehabilitation of the climber with persistent shoulder pain should be based upon management of the underlying physical impairments which have contributed to tissue sensitivity and load intolerance [123]. It is therefore important that the clinician establish the dominate features that are contributing to the climber's symptom provocation [124]. A comprehensive upper quadrant assessment should be undertaken including [125]:

1. Functional demonstration of the activities or climbing movements producing symptoms
2. Contractile function of the rotator cuff musculature and axioscapular muscles (middle trapezius and rhomboids)
3. Soft tissue muscle length of the shoulder adductors and internal rotators (pectoral major, teres major and latissimus dorsi)
4. Glenohumeral joint accessory movement assessment
5. Static upper quadrant postural assessment for muscle balance

An assessment of the trunk musculature, lower quadrant, the neck and its associated nerve plexus may be indicated if there is evidence from the patient's history and/or evidence from the patient's functional climbing demonstration that these factors may be a contributing to shoulder dysfunction. The assessment of the symptomatic regions should be compared to the contralateral side where applicable and used to decide the management approach based upon the impairments present. This approach to the management of shoulder pain is distinctly different from a traditional sports medicine approach of structural tissue failure [126]. It does not attempt to isolate a singular pathological tissue in which to direct medical management. Instead, this approach utilises a holistic approach identifying regions and tissues which are functioning poorly and potentially contributing to ongoing tissue irritation and mechanical sensitivity during high load situations [125]. Rehabilitation interventions are then directed at the areas of failed load tolerance using progressive exercise, adjuncts and electrotherapeutic modalities to restore optimal tissue function and capacity.

20.10 Traumatic Anterior Glenohumeral Dislocation

In first-time traumatic anterior glenohumeral dislocations (FTAGD) in young adults, therapeutic management is often conservative physiotherapy and rehabilitation [127]. Surgery controversial for FTAGD however there are combinations of patient - specific factors such as the presence of meningeal bone loss, significant apprehension and athletes involved in contact sports [128].

Rehabilitation of the FTAGD is predicated upon the stages of soft tissue healing and remodelling (see Table 20.2). The initial stages of rehabilitation should involve appropriate conservative measures to protect the injured tissues from further insult [70]. This will encourage unhindered fibroblastic and proliferation activity. Modalities and adjuncts such as splints and braces to maintain optimal soft tissue length

should be emphasised. Bracing the glenohumeral joint in a shortened position to minimise tensile forces through the injured tissues should be standardised 1–5 days post injury depending upon its severity and degree of associated soft tissue pathology [129]. After a period of initial immobilisation which should be short to prevent complications such as contracture and joint stiffness, a reactivation process should be enacted to allow progressive loading of the injured tissues.

Mechanical loading of the shoulder complex is a cornerstone of rehabilitation at this stage: to reverse arthrogenic muscle inhibition and associated muscle atrophy [130]. It is well accepted that an effusion within or in close proximity to a joint can lead to muscle inhibition [131]. For example, atrophy of the vastus medialis oblique muscle (VMO) in the knee is initiated by minimal effusion. This muscle has similar physiological [132] and morphological [133] characteristics to that of the rotator cuff group in the shoulder. Therefore, muscle activation strategies to preferentially target intramuscular coordination should be instituted as early as possible [133]. Muscle stimulation has been shown to preferentially activate high threshold motor units in atrophied muscles [43].

Isometric muscle training is a safe and effective method to stimulate motor unit recruitment without excessive mechanical load. This is important in the initial 1–2 weeks after the injury because the ligamentous structures are still load compromised [128]. In the middle stage of rehabilitation, higher mechanical loads to stimulate development of connective and muscular tissues is of paramount importance. The mechanical loads should be progressed from isometric to concentric muscle training. Initially, this should be in mid-ranges of motion and positions to lessen the mechanical load on healing structures [134]. This can gradually progress to end of range of motion strengthening in accordance with the climber's functional restrictions, location, direction of apprehension and the mechanism of injury. Low force to higher loading forces is the approach of choice to facilitate appropriate force production and increases in the tissue's material and morphological properties for joint stability [135]. Eccentric loading at a low level can be started



Fig. 20.10 Sensory motor retraining of the glenohumeral joint

with appropriate regard for the stage of soft tissue healing.

Joint stability is dependent upon effective sensory motor control [134]. Injury to the capsule and ligamentous structures affects the afferent feedback, regulation of joint stability and upper quadrant posture and movement [128]. Proprioception offers protection against potentially injurious forces subjected to the shoulder during climbing-related activities [136]. Therefore, sensory motor retraining should be interwoven throughout the rehabilitation after FTAGD (Fig. 20.10). Therapeutic exercise is the primary adjunct. The exercise programme will utilise training exercises that challenge joint position sense in various positions and ranges of motion which are relevant to the climber's return to sport [135]. Late stage rehabilitation will focus on higher level physical conditioning which is specific to climbing. This will include off the wall resis-

tance training modalities to ensure a high degree of tissue robustness [130]. Climbing practice and skill development similar to other rehabilitation modalities starts with low load climbing training and progresses to high load climbing activities. For example, simple climbing problems are progressed to complex climbing problems with high ranges of motion and force demands.

20.11 Elbow

20.11.1 Tendinopathy

The elbow is a common site of pathology in climbers [137]. The largest contributor to this sequelae is tendinopathic pathology [91]. Tendinopathy occurs along a continuum from an acute exacerbation because of abusive mechanical overload due to unaccustomed activity to chronic disrepair and irreversible degenerative changes [138]. The tendons in this region most affected by tendinopathic changes are the common extensor tendons and the common flexor tendons of the elbow [81]. The pathophysiology of tendinopathy is worthy of some consideration because an understanding of the causation will impact the subsequent management. Tendinopathy is associated with pain and disordered collagen within the tendon which lowers its ability to manage mechanical load [139]. It is important to note that tendinopathy represents a non-inflammatory condition with degenerative changes within the body of the tendon and suboptimal collagen synthesis [140]. In this regard, tendinopathy is characterised by a failure of the tendon to remodel after unfavourable mechanical loading. There are several non-mechanical risk factors that can contribute to an impaired tissue response, but this is beyond the scope of this chapter. Tendons are designed to withstand mechanical forces, and appropriate loading has been shown to produce favourable adaptations [141].

The management principles for tendinopathy in the elbow region can be applied to both the common extensor tendons and the common flexor tendons [142]. The point on the continuum

where the climber resides determines the management approach.

In the acute or reactive stage, the tendon has been subject to acute overload or microtrauma, and inflammatory cytokines may be present because of the need for matrix synthesis and degradation [143]. There is a concomitant increase in proteoglycans which attract water and promote changes in the tendon matrix. This has been shown to cause a thickening of the tendon which may or may not serve to reduce further stress and allow some degree of adaptation [144]. However, this thickening is a different response from what would be expected from normal progressive loading. Normal progressive loading results in a stiffer tendon with less thickening [145]. The management of climbers in this stage is one of immediate load reduction. Palliative measures such as the PRICE protocol in the initial stages and taping techniques to offload the pain-sensitive structures are indicated [122, 146]. The reactive tendon then requires a period of slow and progressive mechanical loading over 1–3 weeks. Care must be taken to identify the causative factors that led up to the onset of symptoms, and the climber should be counselled on prevention strategies.

The disrepair stage is a progression of the reactive stage in that the tendon has attempted to heal, but the degree of extracellular matrix degradation exceeds adaptation [147]. This stage is characterised by an increase in fibroblasts and chondrocytes resulting in enhanced proteoglycan production. This leads to a reduction in loading capacity because of disorganisation of the matrix and displacement of the collagen fibres. There is also evidence of neovascularization and neuronal ingrowth [148].

Clinical and empirical evidence appears to suggest that resistance training modalities that promote appropriate mechanical loading consistently perform well [17, 149]. However, no loading scheme (e.g. eccentric, isometric or concentric) has been consistently shown to be superior to another loading scheme. However, there is some evidence that eccentric loading is efficacious when compared to other non-exercise modalities such as ice and therapeutic ultrasound



Fig. 20.11 Resistive training of the forearm extensors using a functional grip position

[150]. There is limited evidence that eccentric loading is superior to any other loading scheme [46]. Therefore, we recommend a combined concentric—eccentric training approach with due regard for symptom provocation. The tendon should be loaded with isolated wrist flexion or extension exercises (dependent upon the site of injury) which cause muscular fatigue at 8–10 repetitions (Fig. 20.11). Using this repetition maximum theorem equates to approximately 75–82% of the muscle's maximum capacity before failure [151]. This degree of loading is appropriate for improving contractile function, reducing force deficits and improving the material and morphological properties of tendon [152].

The training programme should be progressive and adapted to the needs of the climber's functional goals [153]. For example, a bouldering specialist will have greater need for strength, power and shock loading energy storage training [22] during their rehabilitation than an alpine climber. The alpine climber will need tendons that are conditioned to withstand long periods of lower demand loading because of the sport's high endurance demands [8]. Extracorporeal shock wave therapy is an evidence-based modality that has shown efficacy in the management of tendon related disorders [41, 56]. In the climber with a

protracted history of persistent pain and reduced function, this modality can be beneficial as an adjunct to exercise-based training interventions [58].

Acknowledgments We would like to thank our models, Bison Grips, Compex UK, Swiss Dolorclast and the Depot Climbing Centre Birmingham UK for their assistance with this chapter.

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